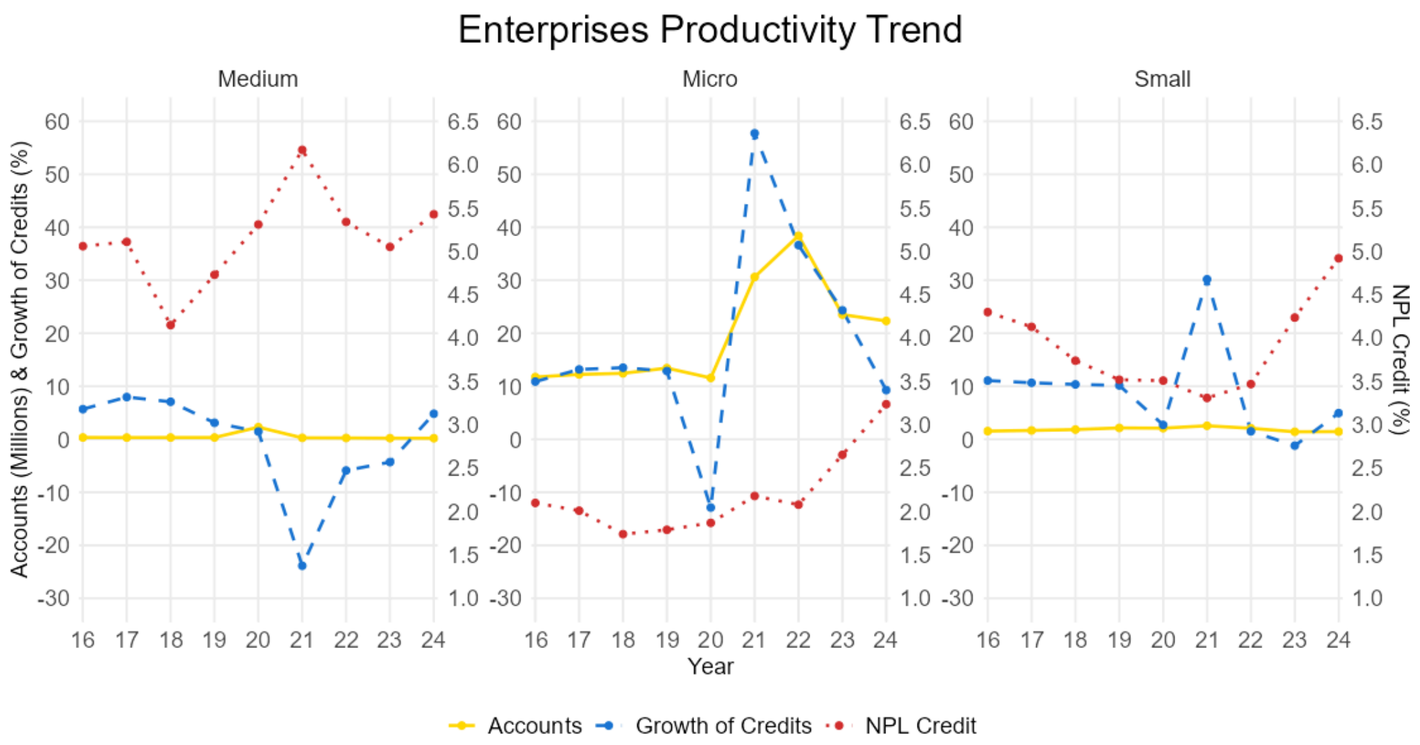
Visualization Project:

Credit Realization Trends of MSMEs by Enterprises Scale and Industry, with Insight into Financial Inclusion of Indonesia

Saldi

Portfolio link: https://github.com/ssal63014-tech/Visualization-Project-Using-R-Shiny



MSMEs Trend in Indonesia

The graph provides a detailed overview of the productivity trends for Medium, Micro, and Small enterprises between 2016 and 2024, focusing on three key indicators: the number of accounts (in millions), the growth of credits (percentage), and non-performing loans (NPL) credit (percentage). Medium enterprises show relatively stable performance, with the number of accounts remaining consistent and credit growth demonstrating moderate fluctuations. However, from 2018 onwards, the NPL credit begins to rise, peaking in 2021, signaling increasing financial strain as these enterprises face challenges in managing debt. Micro enterprises, on the other hand, reveal a high-risk, high-reward dynamic, with aggressive credit growth peaking dramatically in 2021, followed by a sharp decline. This boom-and-bust cycle is mirrored in the NPL credit, which rises sharply, suggesting that the rapid credit expansion may have led to unsustainable debt levels by 2024.

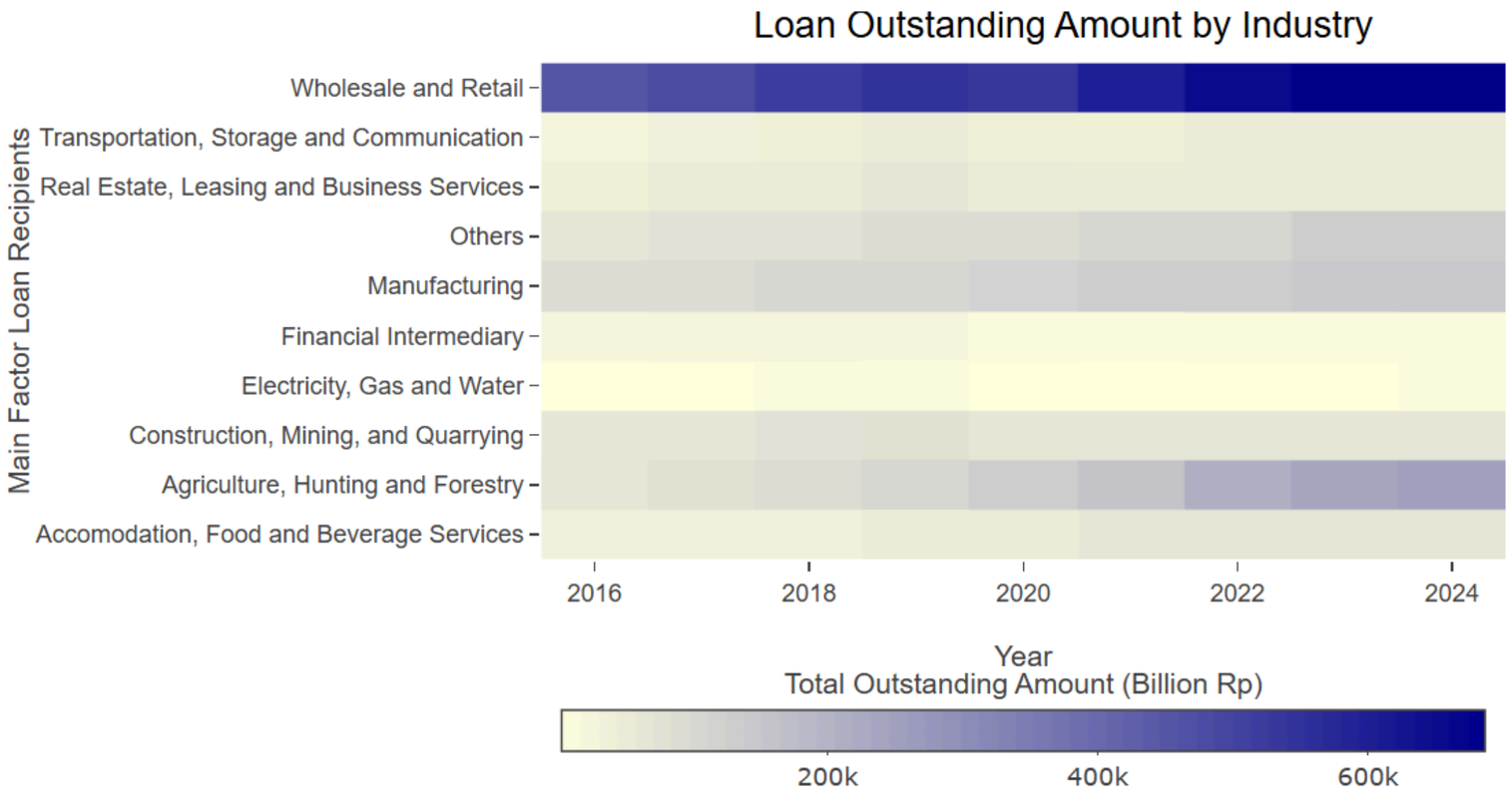
Small enterprises present a more stable trajectory, with the number of accounts remaining relatively consistent, despite fluctuations in credit growth, particularly in 2021. Although NPL credit shows an initial improvement between 2016 and 2021, it starts rising again after 2021, with a significant increase in 2023 and 2024. By 2024, NPL percentages exceed 5%, indicating that many small enterprises struggled with their loan obligations as economic conditions worsened. Overall, the graph reveals key differences in financial health and credit behavior across enterprise categories, highlighting the varying levels of risk and stability within these business segments over time.

Credit Distribution Accross Various Industry

The heatmap bar chart effectively illustrates the outstanding loan amounts across various industries, providing a visual representation that is both informative and intuitive. By hovering over specific areas of the heatmap, users can access detailed information, including the name of the industry, the growth rate of credit, the total outstanding amount for that sector, and the GDP deflator for a given year.

The color gradient in the visualization serves as a key indicator; lighter shades represent lower loan amounts, while progressively darker shades signify higher amounts. Notably, the wholesale and retail industries stand out with the highest outstanding loan amounts over the years, reflecting their significant role in the economy.

Additionally, sectors such as [1] agriculture, hunting, and forestry, [2] manufacturing, and [3] various other industries exhibit a positive trend in credit realization, suggesting a robust expansion in the business market. In contrast, the electricity, gas, and water sectors have shown a concerning decline in credit realization over the same period, highlighting potential challenges within those industries.



Development of Indonesia Financial Inclusion

The visualization of financial inclusion in Indonesia is illustrated through an engaging bubble chart animation, which provides a dynamic overview of the data. Users are invited to explore this interactive tool, where they can click the play button to witness the evolution of financial inclusion over time. This growth is depicted through the interplay of two key factors: the number of service agents and the total number of registered accounts. Each bubble in the chart is color-coded to represent specific regions, while the size of the bubbles corresponds to the total adult population in those regions. This design allows for a comprehensive understanding of how financial services reach different demographics across the country. Users can easily focus on particular regions by selecting them from a checklist, and by hovering over any bubble, they can access detailed information about the number of accounts, service agents, adults, the specific region, and the year of observation.

From the visualization, it becomes evident that Java Island leads the way in financial inclusion. This dominance is directly related to the high number of adults in the region who actively engage with financial services. In stark contrast, Maluku and Papua Islands exhibit a marked lack of financial inclusion growth. This discrepancy can be attributed to their comparatively smaller adult populations—underscoring the challenges these regions face in accessing financial services.

Focusing on the temporal trends, the chart reveals that in 2019, the number of accounts registered for financial services reached its zenith, only to experience a decline in the following year, likely a casualty of the COVID-19 pandemic. Yet, there is a notable rebound in 2021 that indicates an economic recovery, characterized by a diverse distribution of service agents and a renewed growth in registered accounts. This upward trajectory reflects an overall improvement in financial access and utilization as the effects of the pandemic began to wane.

Warner & Spencer

Thank You